Expno: 4 RAW Sockets to implement packet Sniffing

15.10.24

Aim:

To implement packet sniffing using RAW Socket.

Code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
    protocol_name = ""
    if protocol == 1:
        protocol_name = "ICMP"
    elif protocol == 6:
        protocol_name = "TCP"
    elif protocol == 17:
        protocol_name = "UDP"
    else:
        protocol_name = "Unknown Protocol"
```

```
print(f"Protocol : {protocol_name}")
print(f"Source IP : {src_ip}")
print(f"Destination IP : {dst_ip}")
print("-" * 50)

sniff(iface="wifi", prn=packet_callback,
      filter="ip", store=0)
```

Output :

```
Protocol : TCP
Source IP : 57.155.141.117
Destination IP : 192.168.1.6

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Source IP : 192.168.1.6
Destination IP : 57.155.141.117

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Destination IP : 192.168.1.6

Protocol : TCP
Source IP : 192.168.1.6
Destination IP : 57.155.141.117
```

Protocol : UDP
Source IP : 192.168.1.6
Destination IP : 192.168.1.1

Protocol : UDP
Source IP : 192.168.1.6
Destination IP : 192.168.1.1

Protocol : UDP
Source IP : 192.168.1.6
Destination IP : 192.168.1.1.

Result :

Thus, implementation of packet sniffing using RAW socket is done successfully.